

# Thermic Regrading under Norton Multiplication

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## Abstract

Let  $A_u(G) = G.u$  be Norton multiplication of a short partizan game by a positive dyadic number  $u = m/2^k$  in canonical form. Put  $\delta_u = 2^{-k}$ , with  $\delta_u = 1$  for integral  $u$ , and  $a_u = u - \delta_u$ . For every nonnumeric short game,

$$\text{mean}(A_u(G)) = u \text{mean}(G), \quad \text{temp}(A_u(G)) = u \text{temp}(G) + a_u.$$

For a number  $x$  with canonical mesh  $\epsilon_x = 1/\text{den}(x)$ , the image has mean  $ux$  and temperature  $-1$  or  $a_u - u\epsilon_x$ , according as the latter quantity is negative or nonnegative. Hence each positive numeric Norton operator induces an additive transport between thermic residue groups. These transports do not form an action of the multiplicative dyadics: their exact degree defect is

$$\Delta(u, v) = v(1 - \delta_u) - \delta_v + \delta_{uv} \geq 0,$$

and we classify all pairs for which it vanishes. Temperature thus supplies a useful filtered, tropical-like shadow of Norton multiplication, but not a Newton-polygon-style graded ring. The freezing step is not tropical, nonnumeric Norton units do not descend modulo colder games, and the temperature-zero residue retains nonzero 2-torsion.

## 1 Introduction

Temperature is the filtration parameter of thermography: it measures the amount of cooling required before a short game freezes to a number. Norton multiplication  $G.U$  is an additive operation in  $G$  attached to a positive game  $U$  [4, Chapter 8]; modern accounts and related applications appear in [7, 8]. This paper determines exactly how the thermic filtration transforms when the unit  $U$  is a positive number.

The result is affine rather than homogeneous. If  $u$  is a canonical positive dyadic and  $G$  is nonnumeric, then the thermic degree changes by

$$\tau \longmapsto r_u(\tau) = u\tau + u - \delta_u.$$

The denominator mesh  $\delta_u$  is essential. It also produces an exact composition defect, proving that these additive transports cannot be the scalar maps of an associative graded algebra whose dyadic coefficients multiply ordinarily.

The final section compares this result with Newton polygons. The comparison is deliberately limited: both theories have filtrations and extremal piecewise-linear envelopes, but game thermography includes a nontropical freezing operation and has no compatible internal product on Conway values. This sharpens, rather than identifies, the analogy suggested by the standard temperature and valuation formalisms [3, 6].

## 2 Norton and thermographic conventions

We work with short normal-play partizan games modulo Conway equality. Numbers have temperature  $-1$ . For every nonnumber  $G$ , its left and right thermographic walls meet at a finite height  $\text{temp}(G) \geq 0$  and then freeze to the common mast  $\text{mean}(G)$ . We use the standard thermographic conventions of [10, 3].

Norton multiplication depends on the selected form of its positive unit, so throughout a numeric unit is taken in its canonical short-game form. Write a positive dyadic in lowest terms as  $u = m/2^k$  and set

$$\delta_u = \frac{1}{\text{den}(u)} = 2^{-k}, \quad a_u = u - \delta_u. \quad (1)$$

For an integer  $u$ , this gives  $\delta_u = 1$ . A nonintegral canonical dyadic has options  $u - \delta_u$  and  $u + \delta_u$ , while a positive integer has the sole left option  $u - 1$ . Every Norton incentive is consequently  $a_u$ , and the defining recursion becomes

$$A_u(n) = nu \quad (n \in \mathbb{Z}), \quad A_u(G) = \{A_u(G^L) + a_u \mid A_u(G^R) - a_u\} \quad (G \notin \mathbb{Z}). \quad (2)$$

For a number  $x$ , put  $LS_t(x) = RS_t(x) = x$ . For a nonnumber  $G$ , let

$$L_G^{\text{raw}}(t) = \max_{G^L} RS_t(G^L), \quad R_G^{\text{raw}}(t) = \min_{G^R} LS_t(G^R), \quad (3)$$

$$L_t^{\text{sc}}(G) = L_G^{\text{raw}}(t) - t, \quad RS_t^{\text{sc}}(G) = R_G^{\text{raw}}(t) + t. \quad (4)$$

The scaffold walls are followed until their first meeting, after which both are replaced by the common constant mast. The raw gap is

$$D_G(t) = L_G^{\text{raw}}(t) - R_G^{\text{raw}}(t) - 2t. \quad (5)$$

It is nonincreasing, and  $\text{temp}(G)$  is its first zero. We shall also use the standard facts that left walls are nonincreasing, right walls are nondecreasing, and wall slopes lie between  $-1$  and  $1$ .

## 3 The exact thermic law

The recursion stops at integers, not at all numbers. Numeric inputs therefore form a separate branch of the theorem.

**Lemma 3.1** (Numeric inputs). *Let  $x$  be a short-game number and put  $\epsilon_x = 1/\text{den}(x)$ . Then*

$$\text{mean}(A_u(x)) = ux, \quad \text{temp}(A_u(x)) = \begin{cases} -1, & a_u - u\epsilon_x < 0, \\ a_u - u\epsilon_x, & a_u - u\epsilon_x \geq 0. \end{cases}$$

*In particular,  $\text{temp}(A_u(x)) < a_u$ .*

*Proof.* Induct on the exponent of the exact dyadic denominator of  $x$ . Integers are the base case of the Norton recursion. If  $x$  is nonintegral, then  $x = \{x - \epsilon \mid x + \epsilon\}$  with  $\epsilon = \epsilon_x$ . The option images have means  $u(x - \epsilon)$  and  $u(x + \epsilon)$  and, by induction, freeze strictly below  $a_u - u\epsilon$  whenever that value is nonnegative. At  $c = a_u - u\epsilon \geq 0$ , the parent scaffold values are

$$u(x - \epsilon) + a_u - c = ux, \quad u(x + \epsilon) - a_u + c = ux.$$

Immediately below  $c$  their gap is  $2(c - t) > 0$ , so the first meeting is at  $c$  and has mast  $ux$ . If  $c < 0$ , the two numeric bounds in the Norton recursion lie on opposite sides of  $ux$ ; the simplicity rule selects the number  $ux$ .  $\square$

The central wall identity is affine covariance of the raw gap.

**Lemma 3.2** (Raw-gap transport). *Suppose the wall formula*

$$LS_{a_u+ut}(A_u(X)) = uLS_t(X), \quad RS_{a_u+ut}(A_u(X)) = uRS_t(X)$$

*holds for every option  $X$  of a nonintegral game  $G$ . Then*

$$L_{A_u(G)}^{\text{raw}}(a_u + ut) = uL_G^{\text{raw}}(t) + a_u, \quad (6)$$

$$R_{A_u(G)}^{\text{raw}}(a_u + ut) = uR_G^{\text{raw}}(t) - a_u, \quad (7)$$

*and therefore*

$$D_{A_u(G)}(a_u + ut) = uD_G(t). \quad (8)$$

*Proof.* A left option of  $A_u(G)$  is  $A_u(G^L) + a_u$ , and numeric translation shifts both walls by  $a_u$ . Taking the maximum in the definition of the raw left wall proves its displayed transform; the right-option minimum gives the raw right transform. Substitution into the raw-gap definition, with  $s = a_u + ut$ , cancels the two parent shifts and yields the displayed covariance.  $\square$

When  $\text{temp}(G) > 0$ , raw-gap covariance immediately locates the first meeting. The only subtle case is  $\text{temp}(G) = 0 < a_u$ : one must exclude a premature mast below  $a_u$ . We isolate the strategy argument needed for that case.

**Lemma 3.3** (No premature mast). *Let  $X$  be a canonical nonnumber and  $m$  a number. Assume the wall formula of Lemma 3.2 for every proper follower of  $X$ . If, for some  $b < a_u$ ,*

$$RS_s(A_u(X)) = um - a_u + s \quad (b \leq s \leq a_u),$$

*then Right wins  $X - m$  moving first. Dually, if  $LS_s(A_u(X)) = um + a_u - s$  on the same interval, then Left wins  $X - m$  moving first.*

*Proof.* We prove the right-wall statement by birthday induction; negation gives the dual statement. The displayed wall has maximal slope 1, so it is an unfrozen right scaffold on  $[b, a_u)$ . By the Norton recursion,

$$\min_{Y \in X^R} LS_s(A_u(Y)) = um \quad (b \leq s \leq a_u).$$

Choose a right option  $Y$  attaining this minimum at  $b$ . Left walls are nonincreasing and every candidate is at least  $um$  throughout the interval, so  $LS_s(A_u(Y)) = um$  on  $[b, a_u]$ .

If  $Y$  is numeric, Lemma 3.1 shows that it has frozen below  $a_u$  with mast  $uY$ , hence  $Y = m$ . Otherwise, for every left option  $Z$  of  $Y$ , the parent scaffold inequality gives

$$RS_s(A_u(Z)) + a_u - s \leq um.$$

At  $s = a_u$ , the induction hypothesis implies  $RS(Z) \leq m$ . If the inequality is strict, the standard stop criterion says that Right wins  $Z - m$  moving first. If equality holds, the corridor equality excludes numeric  $Z$  by Lemma 3.1: a numeric image is frozen strictly below  $a_u$ , whereas this wall has slope one on a nontrivial interval. The slope bound therefore forces equality throughout the corridor, and birthday induction gives the same conclusion. Right moves from  $X - m$  to  $Y - m$  and answers every left reply using this strategy. Number avoidance shows that a move in the numeric component is no better [10, Part II].  $\square$

**Theorem 3.4** (Thermic regrading). *For every positive canonical dyadic  $u$  and every nonnumeric short game  $G$ ,*

$$\text{mean}(A_u(G)) = u \text{mean}(G), \quad \text{temp}(A_u(G)) = u \text{temp}(G) + a_u. \quad (9)$$

*More strongly, for every  $t \geq 0$ ,*

$$LS_{a_u+ut}(A_u(G)) = uLS_t(G), \quad RS_{a_u+ut}(A_u(G)) = uRS_t(G). \quad (10)$$

*For numeric  $G$ , Lemma 3.1 is the exact replacement.*

*Proof.* Proceed by birthday induction, using Lemma 3.1 for numeric followers. Lemma 3.2 then applies to  $G$ . Put  $\tau = \text{temp}(G)$ . If  $\tau > 0$ , then  $D_{A_u(G)}(a_u) = uD_G(0) > 0$ , so monotonicity excludes an earlier meeting; raw-gap covariance places the first zero at  $a_u + u\tau$ . If  $\tau = 0 = a_u$ , the same identity places it at zero.

Suppose  $\tau = 0 < a_u$ , and write  $LS(G) = RS(G) = m$ . Raw-gap covariance makes the transformed scaffolds meet at  $(a_u, um)$ . If  $A_u(G)$  had frozen at some  $b < a_u$ , its walls would equal  $um$  on  $[b, a_u]$ . For each left option  $X$  of  $G$ , either  $RS(X) < m$ , in which case Right wins  $X - m$  by the stop criterion, or the slope bound forces

$$RS_s(A_u(X)) = um - a_u + s$$

on the corridor, and Lemma 3.3 gives the same conclusion. Thus Right wins  $G - m$  playing second. The dual argument for every right option shows that Left also wins  $G - m$  playing second, whence  $G = m$ , contrary to the assumption that  $G$  is nonnumeric. Premature freezing is impossible.

Before the first meeting, the raw left-wall transform and the scaffold definition give

$$LS_{a_u+ut}^{\text{sc}}(A_u(G)) = uL_G^{\text{raw}}(t) + a_u - (a_u + ut) = uLS_t^{\text{sc}}(G),$$

and similarly on the right. At and above the meeting both sides freeze to the mast  $u \text{mean}(G)$ . This proves both assertions.  $\square$

## 4 Thermic residue transport

Let  $F_{<0} = \mathcal{N}$  be the subgroup of short numbers. For  $\tau \geq 0$ , let  $F_{\leq \tau}$  be the subgroup of games of temperature at most  $\tau$ , including  $F_{<0}$ , and let  $F_{< \tau}$  be the union of lower layers. The standard inequality

$$\text{temp}(G + H) \leq \max\{\text{temp}(G), \text{temp}(H)\} \quad (11)$$

shows that this is an additive filtration [10, Theorem II.5.18].

**Corollary 4.1.** *For every  $\tau \geq 0$ , Norton multiplication by  $u$  induces an additive map*

$$\overline{A}_{u,\tau} : F_{\leq \tau} / F_{< \tau} \longrightarrow F_{\leq u\tau + a_u} / F_{< u\tau + a_u}.$$

*For a positive integer  $n$ , the shifted thermal height  $h(G) = \text{temp}(G) + 1$  satisfies  $h(A_n(G)) = nh(G)$  on nonnumeric layers.*

*Proof.* Norton multiplication by a fixed positive unit is additive [4, 8]. If  $G - H$  has temperature  $\sigma < \tau$ , Theorem 3.4 sends it to temperature  $u\sigma + a_u < u\tau + a_u$ . If  $G - H$  is numeric, Lemma 3.1 puts its image strictly below  $a_u$ . Hence the map is well defined on the quotient. For integral  $n$ ,  $a_n = n - 1$ .  $\square$

## 5 The composition defect

Write  $r_u(\tau) = u\tau + u - \delta_u$ . Successive thermic transports differ from transport by the ordinary dyadic product by a degree-independent term.

**Theorem 5.1.** *For positive dyadics  $u, v$  and every  $\tau \geq 0$ ,*

$$r_v(r_u(\tau)) = r_{uv}(\tau) + \Delta(u, v), \quad \Delta(u, v) = v(1 - \delta_u) - \delta_v + \delta_{uv} \geq 0. \quad (12)$$

*The defect is zero exactly in the following cases:*

1.  *$u$  is nonintegral and  $v = 2^{-\ell}$  for some  $\ell \geq 0$ ;*
2.  *$u$  is an odd positive integer and  $v$  is arbitrary;*
3.  *$u$  is an even positive integer and  $v$  is an integer.*

*Proof.* Expanding the affine maps gives

$$r_v(r_u(\tau)) - r_{uv}(\tau) = v(1 - \delta_u) - \delta_v + \delta_{uv}.$$

If  $u$  is integral, then  $\delta_u = 1$  and  $\Delta = \delta_{uv} - \delta_v$ . Multiplication by an integer cannot increase a dyadic denominator. It preserves every denominator when  $u$  is odd; for even  $u$ , it preserves the denominator exactly when  $v$  is integral. This proves the integral cases and their nonnegativity.

Now let  $u = m/2^k$  in lowest terms with  $k > 0$ . If  $v$  is nonintegral, then  $\delta_{uv} = \delta_u \delta_v$ , and

$$\Delta = (1 - \delta_u)(v - \delta_v) \geq 0.$$

Equality holds precisely when the numerator of  $v$  is 1. If  $v = n$  is an integer, write  $n = 2^s q$  with  $q$  odd. For  $s \geq k$ ,  $\delta_{uv} = 1$  and the defect is positive unless  $n = 1$ . For  $s < k$ ,

$$\Delta = n - 1 - \frac{n - 2^s}{2^k},$$

which is zero at  $n = 1$  and positive for  $n > 1$ . This completes the classification.  $\square$

**Corollary 5.2.** *The residue transports of Corollary 4.1 cannot be the scalar maps of an associative action whose positive dyadic coefficients multiply ordinarily and whose equivalence preserves thermic degree.*

*Proof.* Take  $u = 1/2$ ,  $v = 2$ , and  $G = *$ . Theorem 5.1 gives  $\Delta(1/2, 2) = 1$ , so

$$\text{temp}(A_2(A_{1/2}(*))) = 1, \quad \text{temp}(A_1(*)) = 0.$$

Associativity would identify these two classes, but an equivalence preserving thermic degree cannot do so.  $\square$

## 6 The tropical shadow and its limits

The filtration inequality above is analogous to the nonarchimedean inequality, and the raw option operators are pointwise max-plus and min-plus folds. Cooling adds the affine terms  $-t$  and  $+t$ . This is the exact tropical content of thermography.

The subsequent freeze is different: one compares the two envelopes, finds the first zero of  $D_G$ , and replaces both walls by one constant mast. Nor is the thermograph a congruence for disjunctive sum. The games  $* = \{0 \mid 0\}$  and  $\uparrow = \{0 \mid *\}$  have the same constant-zero walls, but  $* + * = 0$  is numeric while  $\uparrow + *$  is a nonnumber of temperature zero. Thus no binary operation on thermographs alone recovers the thermograph of every sum.

Newton polygons, by contrast, arise from valuations of coefficients and transform functorially under polynomial multiplication; Dumas's theorem is a classical form of this product behavior [6]. The game side has an additive filtration and the external transports of Corollary 4.1, but no analogous internal product on all Conway game values.

The restriction to numeric Norton units is also necessary. At degree zero,  $*$  and  $* + 1$  differ by the cold number 1. For the positive nonnumeric unit  $U = \uparrow$ , additivity gives

$$A_U(* + 1) - A_U(*) = A_U(1) = \uparrow,$$

whose temperature-zero class is nonzero. Thus  $A_U$  does not descend modulo cold numbers. Finding coarser equivalences compatible with Norton multiplication is a separate problem of the kind posed by [2, Problem 6.10].

Finally,

$$\text{gr}_0(\mathcal{G}) = F_{\leq 0}/F_{< 0} \cong \mathcal{I},$$

the additive group of infinitesimals [5, 9]. It contains  $[*] \neq 0$  with  $2[*] = 0$ .

**Proposition 6.1** (Dyadic torsion obstruction). *There is no faithful module containing  $\text{gr}_0(\mathcal{G})$  for which multiplication by a coefficient 2 agrees with game doubling and 2 has a multiplicative inverse representing  $1/2$ .*

*Proof.* Let  $s$  be the image of  $[*]$ . Compatibility with game doubling gives  $2s = 0$ . If 2 is invertible, then  $s = (1/2)(2s) = 0$ , contradicting faithfulness.  $\square$

This obstruction does not exclude integer coefficients, quotients that kill the incompatible torsion, or finer game equivalences. It excludes the specific faithful full-dyadic graded-ring reading.

## 7 Implementation and verification boundary

The Ogdoad implementation contains the standard recursive operations `norton_multiply` and `overheat`, together with three exact closed-form routines:

- `numeric_norton_regrade;`
- `numeric_norton_mean_temperature;`
- `numeric_norton_composition_defect.`

Unit tests compare the closed formulas with recursively materialized products over bounded families of short games and positive dyadic units, and an independent transcription of the Norton recursion checks its noninteger branches. These are regression oracles for the universal proofs above, not their premises. The source is distributed with Ogdoad [1].

The implementation also exposes the tropical folds and the ordinary freeze tail separately. It therefore encodes the same boundary as the paper: the max-plus/min-plus structure belongs to the option envelopes, while the meeting and freezing operation remains ordinary thermography.

## 8 Conclusion

Positive numeric Norton multiplication acts on thermic degree by the exact affine map

$$\tau \longmapsto u\tau + u - \delta_u,$$

with a strictly lower numeric branch. It consequently induces additive maps between thermic residue groups. The nonnegative defect  $\Delta(u, v)$  measures exactly why those maps fail to form a multiplicative dyadic action. The tropical comparison is therefore a filtered one: thermography has extremal envelopes and a nonarchimedean temperature inequality, but its freeze, its nonnumeric Norton operators, and its temperature-zero torsion prevent a faithful Newton-style graded ring.

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